University of Wisconsin-Milwaukee

College of Engineering and Applied Science

Midterm Exam(In Class: 1:00PM - 2:15 PM) 75-Minutes

Wednesday-23rd October – Fall 2024

Course Title: Algorithm Design and Analysis
Instructor: Rohit P. Singh
Course Code: CompSci 535

Total Marks: 125
Panthers- ID : _____
Student's name: _____

Answer the questions in the spaces provided. If you run out of room, continue on the back of the page.
Critical Thinking is a must for algorithm design (There could be multiple solutions to the same problem;)
1: In class Exam — 125 marks
2: The exam is a closed book and notes. No electronic devices are allowed.
3: Watch out for time and use it optimally.
4: All questions are Compulsory. Good Luck.

Q.1) Answer the Following Questions:

(a) (3-Marks) The worst-case time complexity of a search in a binary search tree is $O(log(n))$. The given tree is balanced binary search tree[T/F]._____.

(b) (3-Marks)The height of all external nodes in 2-4 tree is same.[T/F]._____.

(c) (3-Marks) The minimum height of the multi-way search tree with n elements and order k is exactly $\lfloor log_k(n+1) \rfloor$[T/F]._____.

(d) (3-Marks)For a heap with n elements, its minimum and maximum height are different. [T/F]._____.

(e) (3-Marks)The time complexity of in-order, post-order and pre-order traversals of balanced binary search trees with n elements is $O(log(n))$.[T/F]._____.

(f) (3-Marks)The deletion of nodes from binary trees having both children as internal nodes can be accomplished by replacing it with either a successor or a predecessor. The selection of successor or predecessor are both equally good choices [T/F]._____.

(g) (5-Marks)For double hashing with a hash table of size 43. Write down hash functions $h(k)$ and $d(k)$ as a function of k(key) and i(number of collisions) only. _____.

(h) (5-Marks)What is the shape-property of heaps: _____

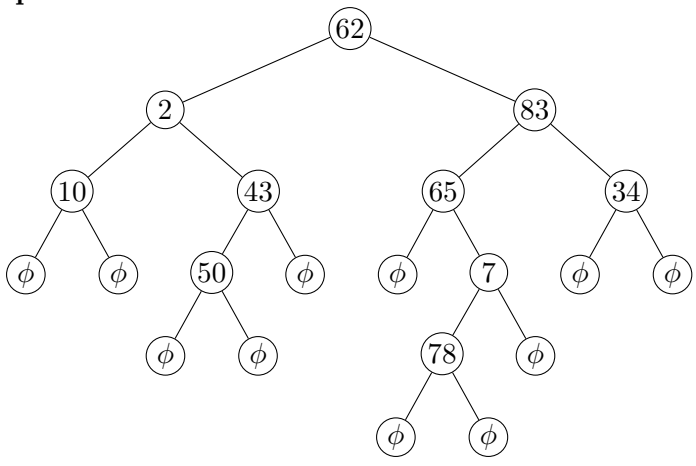
_____.

 Question 1 continues on the next page...

Rough Work

Question 1 continues on the next page...

For a given binary tree below with ϕ as external nodes. Answer the following questions:



(i) (3-Marks) Write down the pre-order traversal of the above tree.

(j) (3-Marks) Write down the in-order traversal of the above tree.

(k) (2-Marks) Write down the post-order traversal of the above tree.

(l) (2-Marks) Write down the level-order traversal of the above tree.

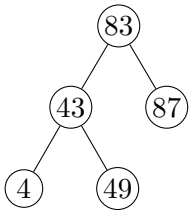
(m) (2-Marks) Is this tree an AVL tree? If so, why? If not, why not?

Total for Question Q.1: 40 marks

Question 1 continues on the next page...

Rough Work

Q.2) Consider the splay tree below:



Insert the following keys in a given order into the above splay tree 24,62 and 52. Show the changes happening to the tree at each step. Total for Question Q.2: 15 marks

Question 2 continues on the next page. . .

Rough Work

Q.3) (a) (5 Marks) (5 Marks)What is hashing? What is the major drawback of linear probing? How do quadratic and double hashing overcome the drawback of linear probing?

(b) (5 Marks) Consider input keys as strings of small-cap alphabets(a-z). The **Integer Code**(IC) for a alphabet is as follows $a = 0, b = 1, c = 2, d = 3, e = 4, \ldots z = 25$ and can be retrieved by IC(alphabet). For example $IC(a) = 0$ and $IC(f) = 5$.

Write a pseudo-code for a function HASH-CODE-MAPS that takes key k of type string and integer constant z as input. The hash-code map returns $p(k, z)$ as given below:

$p(k, z) = IC(k[0]) + IC(k[1])z + IC(k[2])z^2 + \ldots + IC(k[len(k) - 1])z^{len(k)-1}$

```
int IC(char x){
    return x-97;
}
int HASH-CODE-MAPS(string k,const int z){

}
```

Rough Work

Question 3 continues on the next page. . .

(c) (5 Marks)Write a pseudo-code to INSERT a valid key k into a hash table T of size N that handles a collision with linear probing. You can directly use the function HASH-CODE-MAPS developed above to map keys to integers.

```
void INSERT(string k,const int z){

}
```

(d) (5 Marks)Insert the following keys $\{aa, bc, df, ega, hb, aaa\}$ into a hash table of size $N = 7$ using INSERT function developed above. Use $z = 1$. Show the hash table status and the number of collision for each insertion.

Total for Question Q.3: 20 marks

Question 3 continues on the next page...

Rough Work

Q.4) Asymptotic Complexity

(a) (15 Marks) For each of the items below, indicate whether $f(n)$ is $\Omega(g(n))$, $\Theta(g(n))$ or $O(g(n))$. Show how you arrived at your answer.

1. $f(n) = 8^{\frac{n}{3}}$ and $g(n) = 16^{\frac{n}{4}}$
2. $f(n) = 1 + 2 + 3 + 4 \ldots + n^2$ and $g(n) = n^2 + \frac{n^2}{2} + \frac{n^2}{4} + \ldots + \frac{n^2}{2^{logn}}$
3. $f(n) = 8^{logn}$ and $g(n) = (n + n^2)^2$
4. $f(n) = 4n^2 + 6n + 7$ and $g(n) = 105n^2 + 1$

Question 4 continues on the next page...

Rough Work

Question 4 continues on the next page. . .

(b) (10-Marks)Explain Asymptotic and Amortized time complexities. Find the running time of the following code snippets with input variable n. Show how you arrived at the results.

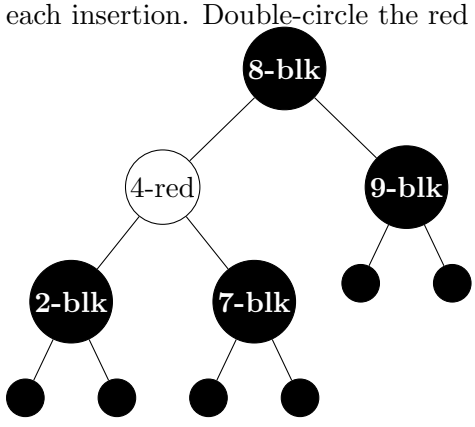
```
for( int m = n; m > 0; m /= 2 ) {
     for( int q = 1; q < n; q *= 2 ) {
            for( int p = 0; p < n; p += 2 ) {
                ----constant time operation
            }
     }
}
```


Question 4 continues on the next page. . .

Rough Work

Total for Question Q.4: 25 marks

Q.5) (a) (10 Marks) Insert 45, 23 and 25 into the following Red Black Tree. Show the tree after each insertion. Double-circle the red nodes or annotate them with a red or black label.

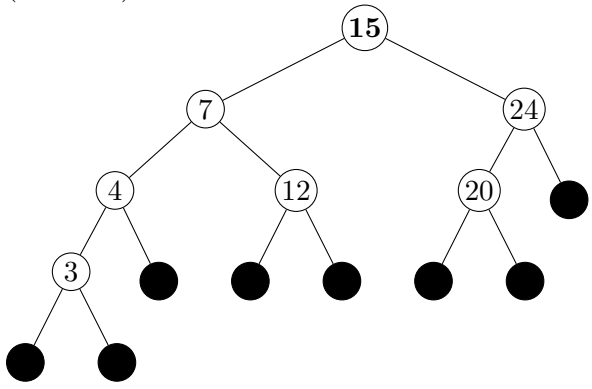


Question 5 continues on the next page. . .

Rough Work

Question 5 continues on the next page...

(b) (5 Marks)Delete 15 and then 12 from following AVL Tree. Show the tree after each deletion.



(c) (10 Marks)Write a pseudo-code to verify whether a given binary search tree is AVL.

Total for Question Q.5: 25 marks

Question 5 continues on the next page...

Rough Work

Question 5 continues on the next page...

Question	Points	Score
Q.1	40	
Q.2	15	
Q.3	20	
Q.4	25	
Q.5	25	
Total:	125	